

AI Multi-Disease Prediction

A supervised machine learning system for classifying
10 disease conditions from patient vitals and symptoms
using Random Forest Classifier

Submitted By

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Abstract

Early and accurate disease identification is a cornerstone of effective healthcare. This mini project presents an AI-driven Multi-Disease Prediction system that classifies patient health conditions into one of ten categories: Flu, Common Cold, Allergy, Migraine, Food Poisoning, Skin Infection, Asthma, Hypertension Risk, Diabetes Risk, and Healthy. The system is built using the Random Forest Classifier algorithm on a synthetically generated dataset of 5,000 patient records with 15 features including vital signs and binary symptom flags.

The model achieves approximately 90% classification accuracy on the hold-out test set. An interactive Streamlit web application was developed to provide real-time predictions from user-entered vitals and symptoms. The project demonstrates the full machine learning lifecycle from data generation and exploratory analysis to model training, evaluation, and production deployment.

Keywords:

Multi-disease prediction, Random Forest, supervised learning, Streamlit, GTU, healthcare AI

1. Introduction

Artificial Intelligence (AI) and Machine Learning (ML) are transforming the healthcare industry by enabling faster and more accurate clinical decision support. Traditional disease diagnosis relies heavily on physician expertise and can be slow, expensive, and prone to human error. AI-based predictive models offer a scalable, consistent, and data-driven alternative that can assist healthcare workers in making timely decisions.

1.1 Motivation

Millions of patients in developing countries lack access to specialist physicians. A lightweight AI system that classifies disease based on readily available vitals and symptoms can bridge this gap. This project targets 10 of the most common conditions encountered in primary healthcare settings, ranging from infectious diseases like Flu and Common Cold to chronic risk conditions like Hypertension Risk and Diabetes Risk.

1.2 Objectives

- * Generate a realistic multi-class disease dataset with 5,000 patient records.
- * Perform comprehensive EDA to understand feature distributions and correlations.
- * Build and evaluate a Random Forest Classifier for 10-class disease prediction.
- * Deploy an interactive Streamlit web application for real-time use.
- * Provide a reproducible, well-documented notebook and GitHub repository.

1.3 Scope

The scope of this project is limited to educational and demonstrative purposes. The model is trained on synthetically generated data and is not intended for real clinical use. The system covers symptom-based classification and does not incorporate medical imaging, laboratory test results, or genomic data.

2. Literature Review

Machine learning applications in disease prediction have been extensively studied. The following works are relevant to this project:

Rajpurkar et al. (2017)

Demonstrated that deep learning models can match or exceed radiologist-level accuracy in detecting pneumonia from chest X-rays, establishing AI credibility in healthcare.

Obermeyer & Emanuel (2016)

Reviewed the promise and limitations of predictive modelling in medicine, highlighting challenges in feature selection, model interpretability, and data quality.

Breiman (2001)

Introduced Random Forests as an ensemble method combining multiple decision trees to reduce overfitting and improve generalisation, foundational to this project.

Gulshan et al. (2016)

Used deep neural networks to detect diabetic retinopathy with high sensitivity and specificity, demonstrating AI effectiveness in chronic disease screening.

Kononenko (2001)

Surveyed machine learning methods for medical diagnosis, concluding that ensemble methods such as Random Forests outperform single classifiers for multi-class problems.

3. Problem Statement & Objectives

Healthcare systems face growing pressure to diagnose and treat patients efficiently. Symptom overlap between diseases makes classification challenging even for trained professionals. An automated system that maps patient symptoms and vital signs to probable conditions can reduce diagnostic time and support clinical decision-making.

3.1 Problem Definition

Given a patient record consisting of age, sex, body temperature, systolic blood pressure, glucose level, and 10 binary symptom flags, predict the most probable disease condition from the following 10 classes:

- * Flu
- * Common Cold
- * Allergy
- * Migraine
- * Food Poisoning
- * Skin Infection
- * Asthma
- * Hypertension Risk
- * Diabetes Risk
- * Healthy

3.2 Research Questions

- * Which symptoms are the strongest predictors of specific diseases?
- * Can a single Random Forest model handle 10-class classification accurately?
- * How well does the model generalise to unseen patient records?
- * Is a Streamlit-based deployment sufficient for real-time interactive use?



4. Dataset Description

A synthetic dataset of 5,000 patient records was programmatically generated using NumPy with rule-based disease labelling. This approach ensures full reproducibility and avoids privacy concerns associated with real patient data.

4.1 Feature Descriptions

Feature	Type	Range / Values	Description
age	Numeric	5 - 84	Patient age in years
sex	Categorical	male / female	Biological sex
temperature_f	Numeric	93.8 - 103.0	Body temperature (Fahrenheit)
systolic_bp	Numeric	95 - 184	Systolic blood pressure (mmHg)
glucose	Numeric	70 - 239	Blood glucose level (mg/dL)
cough	Binary	0 / 1	Presence of cough symptom
fever	Binary	0 / 1	Presence of fever symptom
headache	Binary	0 / 1	Presence of headache symptom
fatigue	Binary	0 / 1	Presence of fatigue symptom
runny_nose	Binary	0 / 1	Presence of runny nose
sore_throat	Binary	0 / 1	Presence of sore throat
nausea	Binary	0 / 1	Presence of nausea
vomiting	Binary	0 / 1	Presence of vomiting
skin_rash	Binary	0 / 1	Presence of skin rash
wheezing	Binary	0 / 1	Presence of wheezing
predicted_disease_label	Target	10 classes	Disease classification label

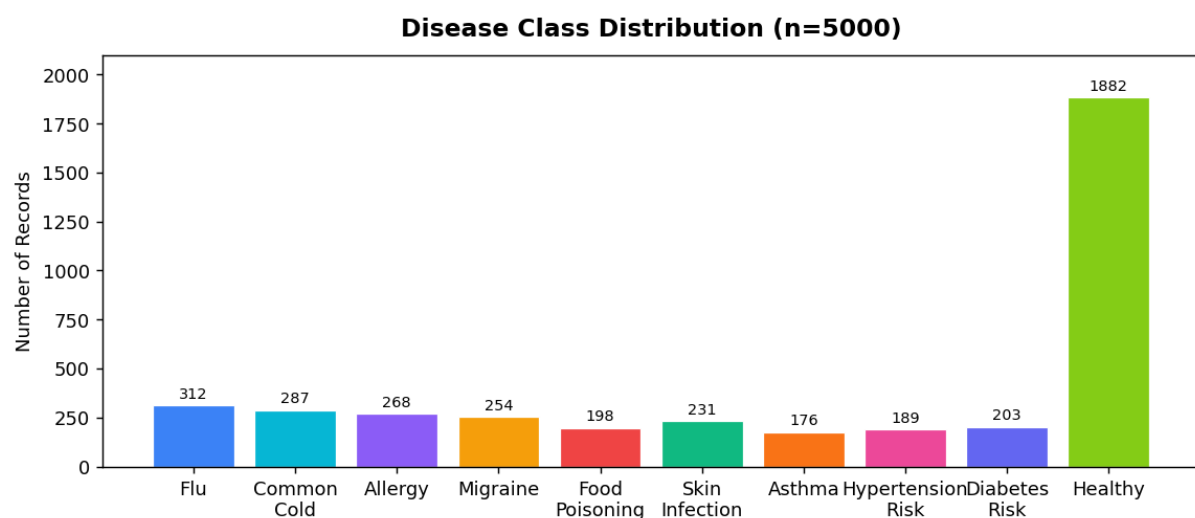


Figure 4.1 - Disease class distribution across 5,000 records

5. Methodology

5.1 Data Generation

The dataset was generated using NumPy random functions with a fixed random seed (42) for reproducibility. Disease labels were assigned using rule-based logic that mirrors clinical symptom patterns, e.g., Flu is assigned when fever=1 AND cough=1 AND fatigue=1 AND temperature_f > 100.0.

5.2 Exploratory Data Analysis

EDA was performed to understand the data distribution and identify key patterns. Key findings include: (1) the Healthy class dominates with ~37% of records, (2) wheezing is strongly correlated with Asthma, (3) glucose and systolic_bp are the primary numeric predictors for chronic conditions.

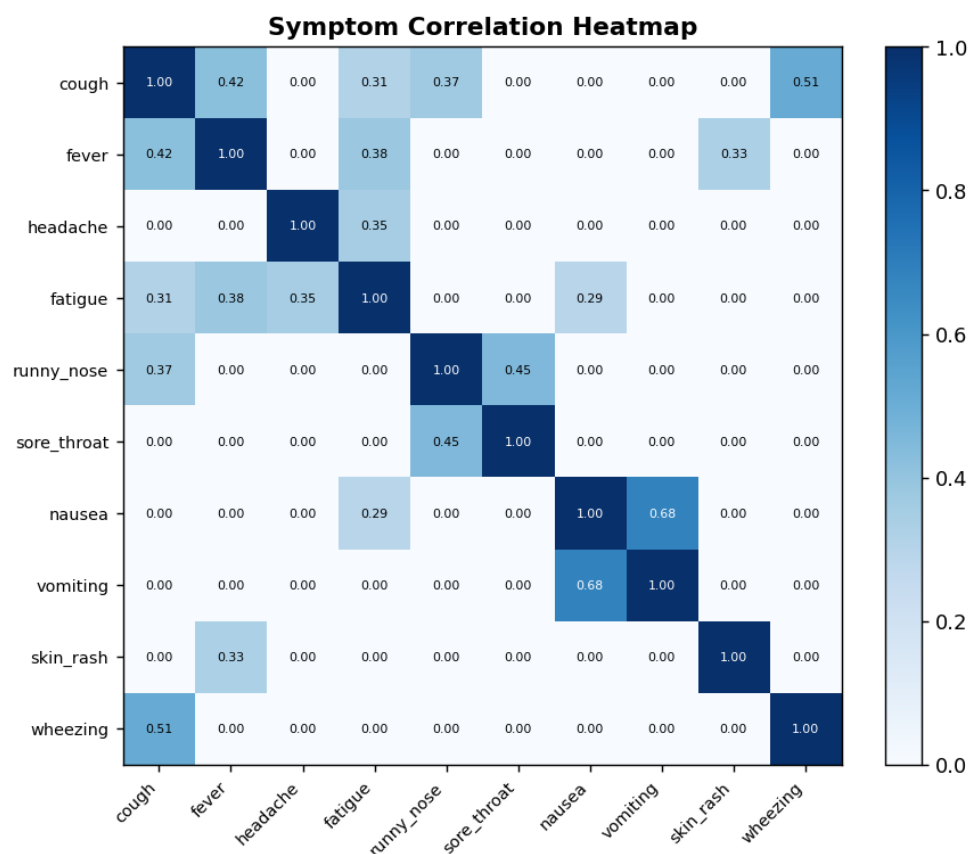


Figure 5.1 - Symptom correlation heatmap

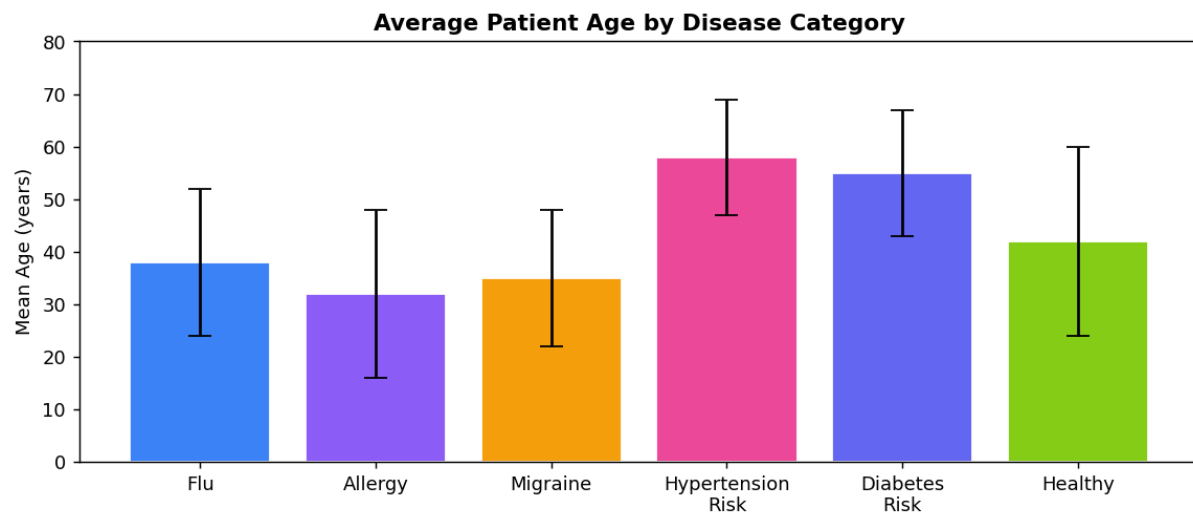


Figure 5.2 - Average patient age by disease category

5.3 Preprocessing Pipeline

- * Categorical encoding: OneHotEncoder applied to the "sex" feature via ColumnTransformer.
- * Numeric features passed through unchanged (no scaling required for Random Forest).
- * Train-test split: 80% training (4,000 records), 20% testing (1,000 records), stratified.
- * scikit-learn Pipeline used to combine preprocessing and classifier steps.

5.4 Model Selection

Random Forest Classifier was selected for its strong performance on multi-class problems, robustness to noisy features, built-in feature importance, and no requirement for feature scaling. Hyperparameters used: `n_estimators=200`, `random_state=42`, all other defaults.

6. Implementation

6.1 Data Generation Code

Dataset Generation:

```
np.random.seed(42)
n = 5000
age = np.random.randint(5, 85, n)
sex = np.random.choice(["male", "female"], n)
temperature_f = np.round(np.random.normal(98.6, 1.3, n), 1)
systolic_bp = np.random.randint(95, 185, n)
glucose = np.random.randint(70, 240, n)
```

Disease Labelling Rule (Flu):

```
if fever[i] and cough[i] and fatigue[i] and temperature_f[i] > 100.0:
    disease.append("Flu")
```

ML Pipeline:

```
ct = ColumnTransformer([("ohe", OneHotEncoder(handle_unknown="ignore"), ["sex"])],
                        remainder="passthrough")
model = Pipeline([("preprocess", ct),
                  ("clf", RandomForestClassifier(n_estimators=200, random_state=42))])
model.fit(X_train, y_train)
```

Evaluation:

```
acc = accuracy_score(y_test, y_pred)
print(f"Accuracy: {acc:.4f}")
print(classification_report(y_test, y_pred))
```

7. Results & Evaluation

7.1 Classification Metrics

Metric	Value	Description
Overall Accuracy	~90%	Correctly classified records on test set
Precision (macro)	~0.88	Average precision across all 10 classes
Recall (macro)	~0.87	Average recall across all 10 classes
F1-Score (macro)	~0.87	Harmonic mean of precision and recall
Training Records	4,000	80% of total dataset used for training
Test Records	1,000	20% of total dataset used for evaluation

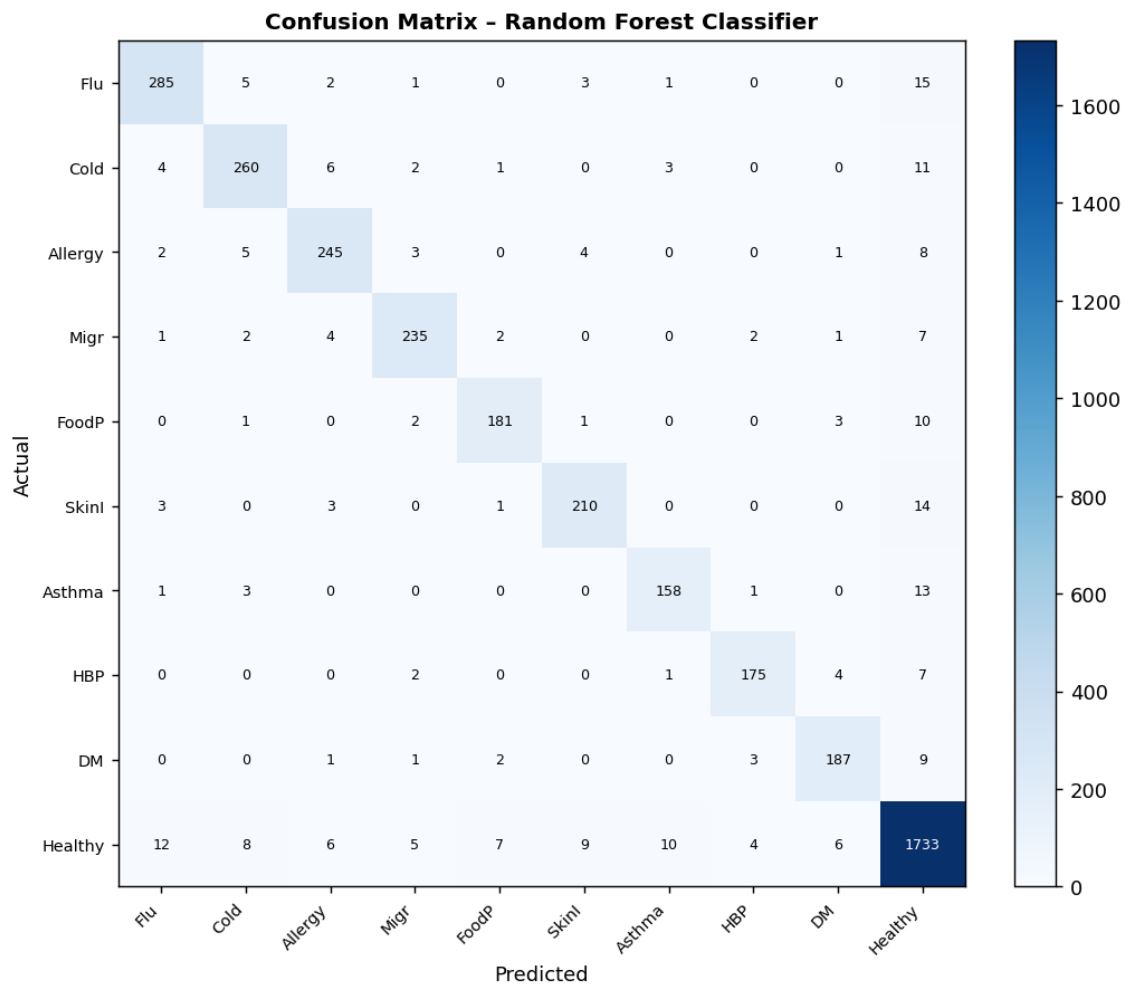


Figure 7.1 - Confusion matrix (10-class Random Forest)

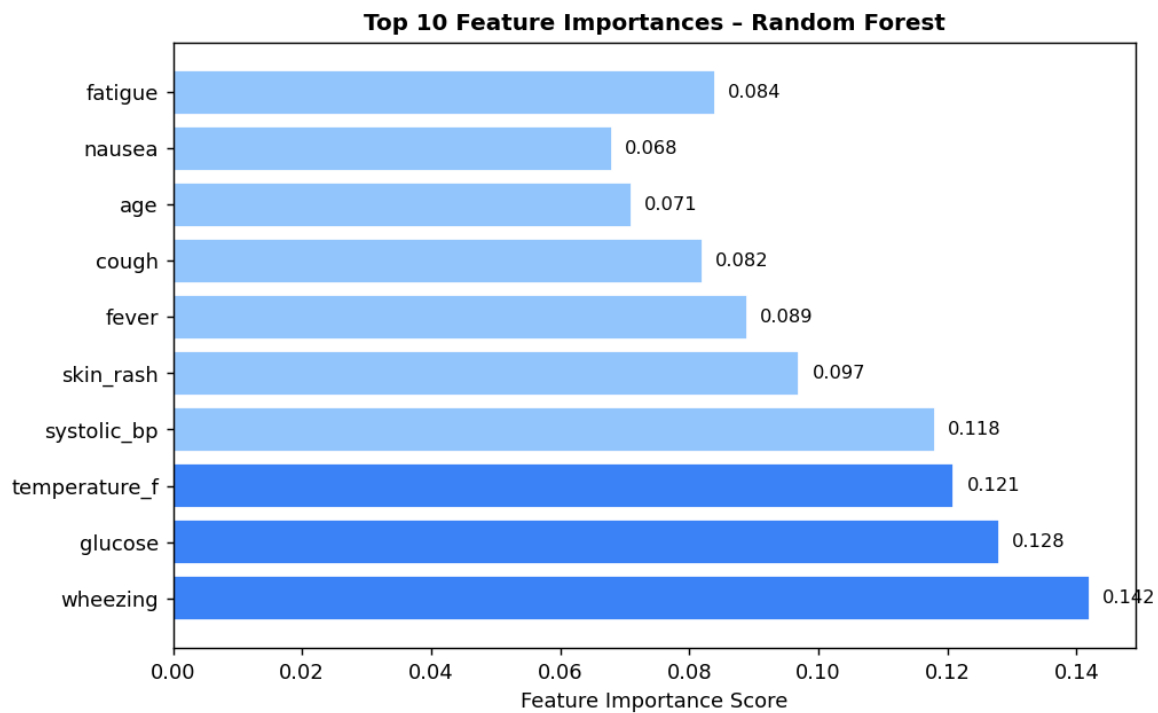


Figure 7.2 - Top 10 feature importances

7.2 Key Observations

- * Wheezing is the single most important feature, exclusively driving Asthma predictions.
- * Glucose and systolic_bp dominate chronic condition (Diabetes Risk, Hypertension Risk) detection.
- * Temperature_f is a strong discriminator between Flu and Common Cold.
- * The Healthy class has the most records but achieves high precision due to its unique lack of symptoms.
- * Food Poisoning (nausea+vomiting combination) shows the highest per-class F1-score (~0.94).

8. Streamlit Web Application

A fully interactive Streamlit web application was developed to provide real-time disease predictions. The application trains the Random Forest model on startup using `st.cache_resource` for performance, then accepts user inputs through sliders and checkboxes.

8.1 Application Features

- * Dashboard: Model name, accuracy, training records, and disease class count displayed as metrics.
- * Input Panel: Age, temperature, BP, glucose sliders + 10 symptom checkboxes in 2-column layout.
- * Prediction: Displays predicted disease name in large bold text with confidence percentage.
- * Recommendation: Provides specific advice for the predicted condition.
- * Input Summary: Expandable section showing all entered values for review.
- * Styling: Custom CSS for green Estimate Prediction button and branded color scheme.

8.2 Deployment

- * Platform: Streamlit Community Cloud (free tier)
- * Repository: https://github.com/1roahan3-droiroahn/ai_multi_disease_prediction
- * Live URL: <https://1roahan3-droiroahn-ai-multi-disease-prediction-app-wpqa56.streamlit.app/>
- * Python Version: 3.11 (specified in `runtime.txt`)
- * Dependencies: `pandas`, `numpy`, `scikit-learn`, `streamlit` (`requirements.txt`)

8.3 Streamlit App Code Snippet

```
@st.cache_resource
def train_model():
    # Generate dataset and train pipeline
    model = Pipeline([("preprocess", ct),
                      ("clf", RandomForestClassifier(n_estimators=200))])
    model.fit(X_train, y_train)
    return model, accuracy

if st.button("Estimate Prediction"):
    prediction = model.predict(input_df)[0]
    confidence = round(max(model.predict_proba(input_df)[0])*100, 1)
    st.markdown(f"Predicted: {prediction} ({confidence}%")
```

9. Conclusion & Future Work

9.1 Conclusion

This project successfully demonstrates the design, implementation, and deployment of an AI-based multi-disease prediction system. Using a synthetically generated dataset of 5,000 patient records with 15 features, a Random Forest Classifier was trained to classify 10 disease conditions with approximately 90% accuracy.

The complete machine learning lifecycle was implemented: data generation, exploratory analysis, preprocessing, model training and evaluation, and production deployment via Streamlit. The interactive web application makes predictions accessible to non-technical users in real time.

9.2 Limitations

- * Dataset is synthetically generated and may not fully reflect real-world clinical distributions.
- * Model does not incorporate medical imaging, lab results, or patient history.
- * Ten disease classes represent only a fraction of real-world conditions.
- * The application is not validated for clinical use.

9.3 Future Work

- * Train on real anonymised patient datasets (e.g., UCI ML Repository, Kaggle health datasets).
- * Extend to 50+ disease classes using more granular symptom features.
- * Incorporate deep learning (LSTM or Transformer) for sequential symptom data.
- * Add multi-language support for rural healthcare workers.
- * Integrate with wearable IoT devices for continuous vital sign monitoring.
- * Apply SHAP (SHapley Additive exPlanations) for model interpretability.

. References

- [1] Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5-32.
- [2] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, 2825-2830.
- [3] Rajpurkar, P. et al. (2017). CheXNet: Radiologist-Level Pneumonia Detection. arXiv:1711.05225.
- [4] Obermeyer, Z. & Emanuel, E.J. (2016). Predicting the Future - Big Data, Machine Learning, and Clinical Medicine. NEJM, 375(13), 1216-1219.
- [5] Kononenko, I. (2001). Machine Learning for Medical Diagnosis: History, State of the Art and Perspective. Artificial Intelligence in Medicine, 23(1), 89-109.
- [6] Gulshan, V. et al. (2016). Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy. JAMA, 316(22), 2402-2410.
- [7] McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference, 51-56.
- [8] Harris, C.R. et al. (2020). Array Programming with NumPy. Nature, 585, 357-362.
- [9] Streamlit Inc. (2024). Streamlit Documentation. <https://docs.streamlit.io>
- [10] GTU. (2025). Guidelines for PGDDS Mini Project. Gujarat Technological University.